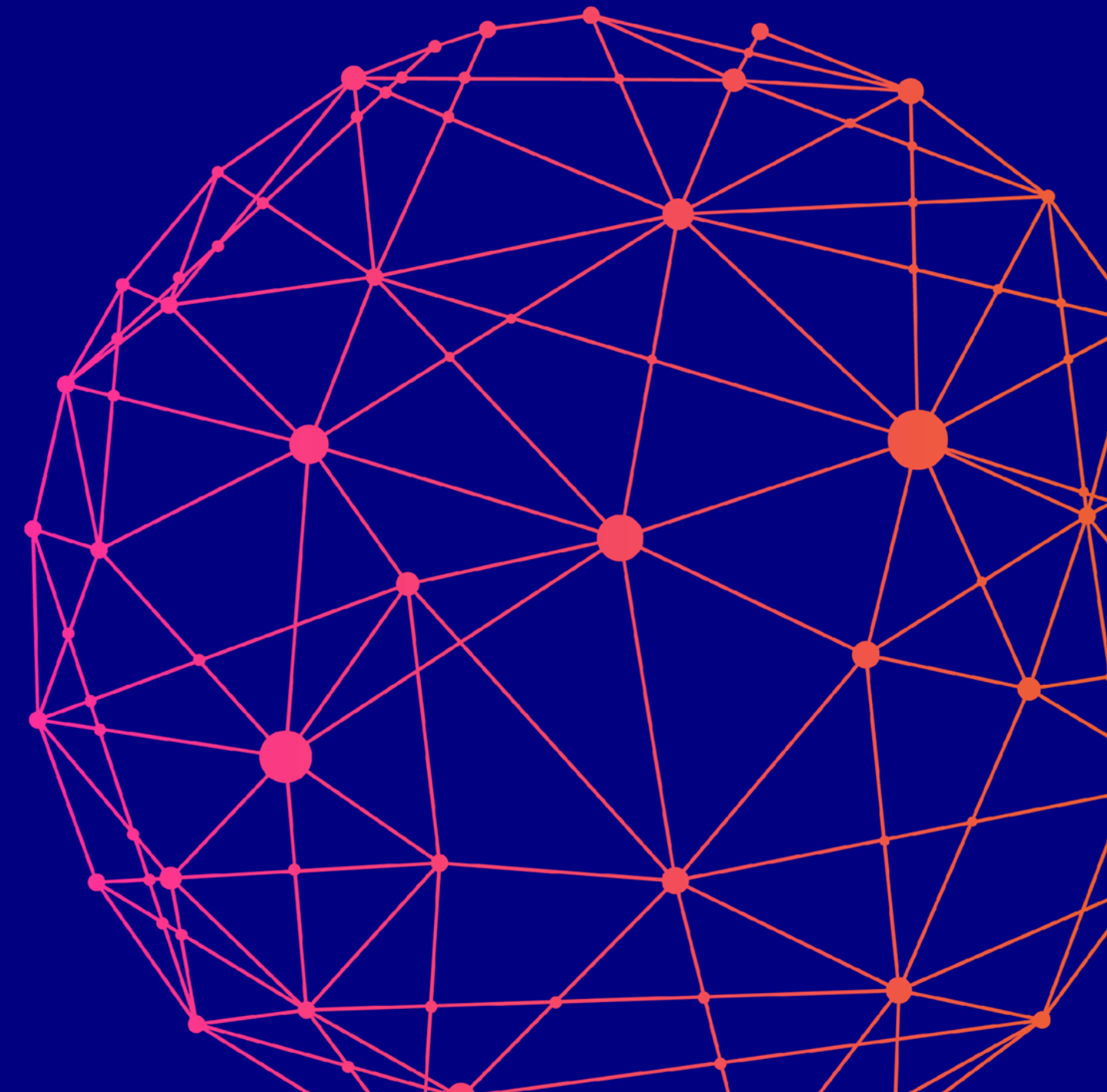


Data Spaces Symposium

Accelerating adoption. Increasing impact.

Empowering Interoperability:
EDC and Framework
Essentials for Data Spaces



Data Spaces Symposium

Speaker Intro



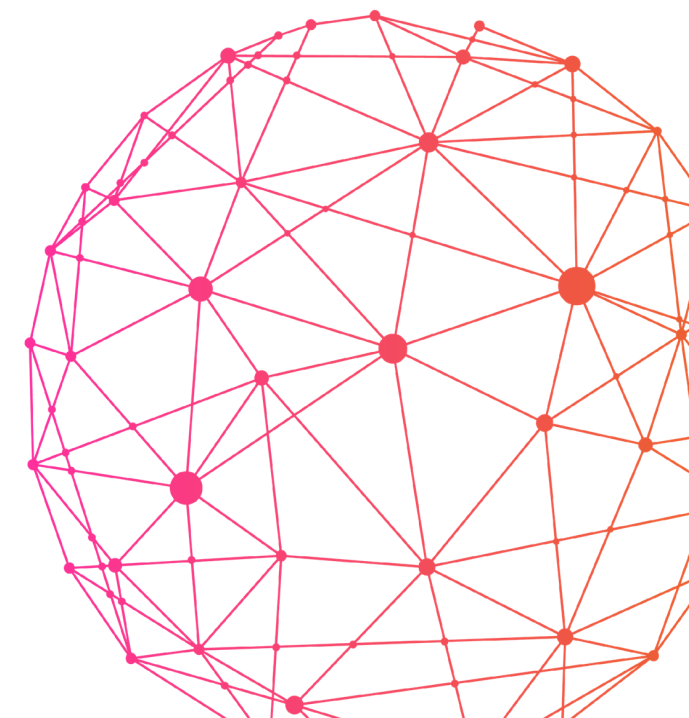
Pol López



PwC's Data Spaces Offering:

- Trusted neutral orchestrator in data ecosystems
- Deep data-space expertise across strategy, governance & architecture
- Unlock business value through scalable, cross-industry use cases

Explore our
Centre of
Excellence for
Data Spaces →



Data Spaces Symposium

Towards Dataspace Implementations

Motivation:

- Focus on **business opportunities** and **revenue generation**
- Ramp-up technology for the **data economy**, esp. (agentic) **AI**

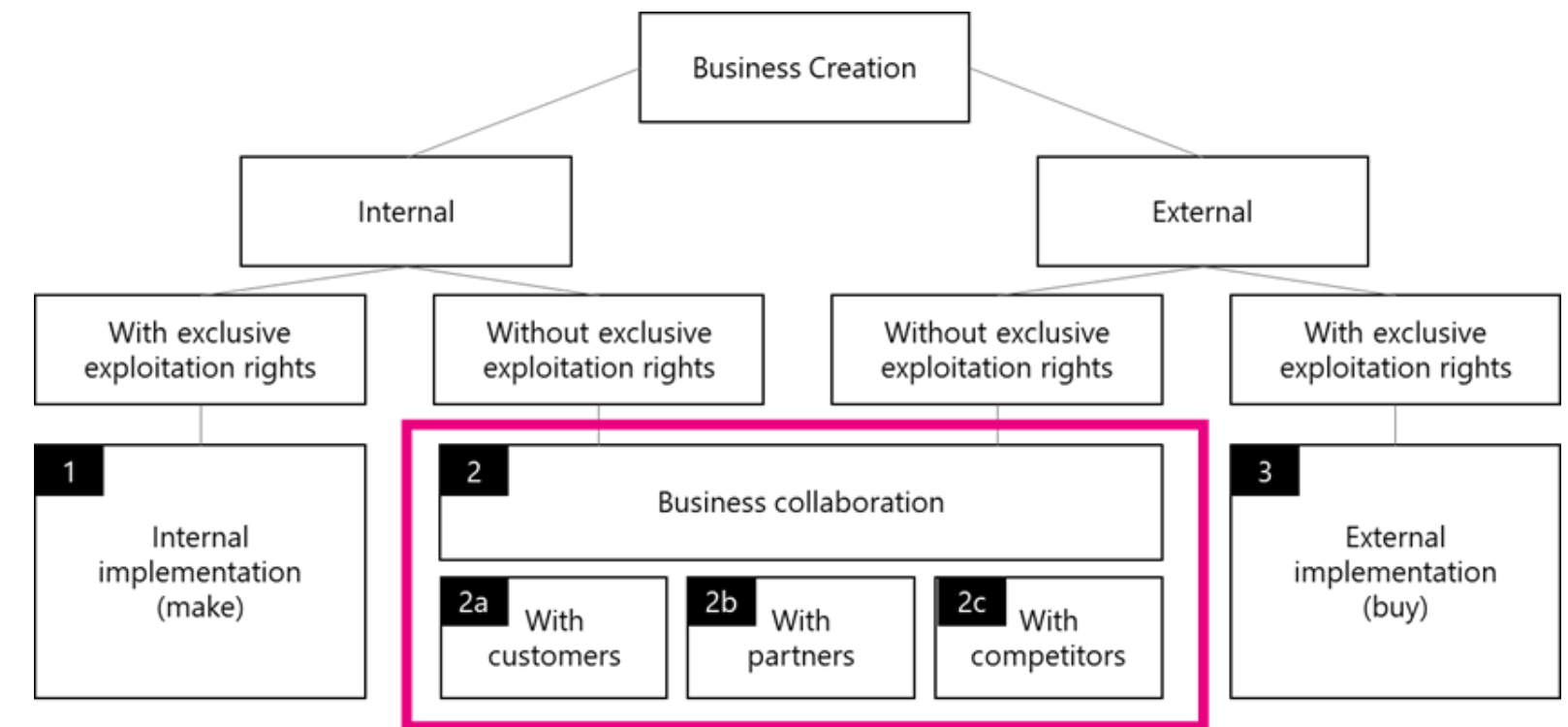
Requirements:

- Common (technical) protocols
- Clear architecture & reference implementation(s)
- Infrastructure portability (on-prem & cloud)
- Open for (regional or domain-specific) customization

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The Role of Open Source in the Dataspace Domain

- Based on open, international standards
- Requires high level of adoption
- Serve a range of use cases and requirements
- Require coherence on technical layer
- Allow flexibility for business and legal layers
- Infrastructure & provider agnostic

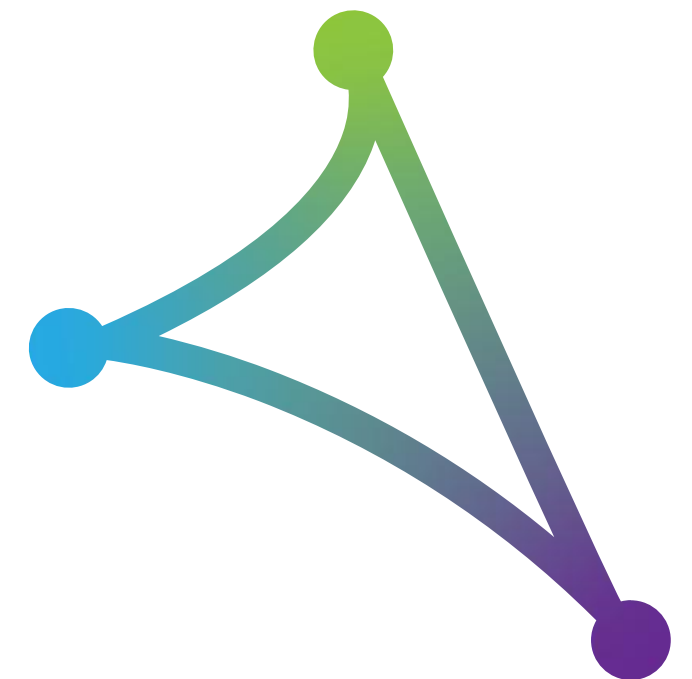


By working with open source to build dataspace, we can **achieve more together!**

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Framework to Build Dataspace Components

- Eclipse Dataspace Components (EDC)
- Project governed by the Eclipse Foundation
- Open, transparent, meritocratic, vendor neutral
- Licensed under Apache 2.0 License
- Baked by leading companies (80+ contributors)
- Adopted by more than 40 dataspace initiatives
- Reference implementation of dataspace protocols



EDC was designed with the philosophy that **one size does not fit all**. Before deploying an EDC-powered data sharing ecosystem, you'll need to build **customizations** and bundle them into **one or more distributions**.

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Foundational Guidelines

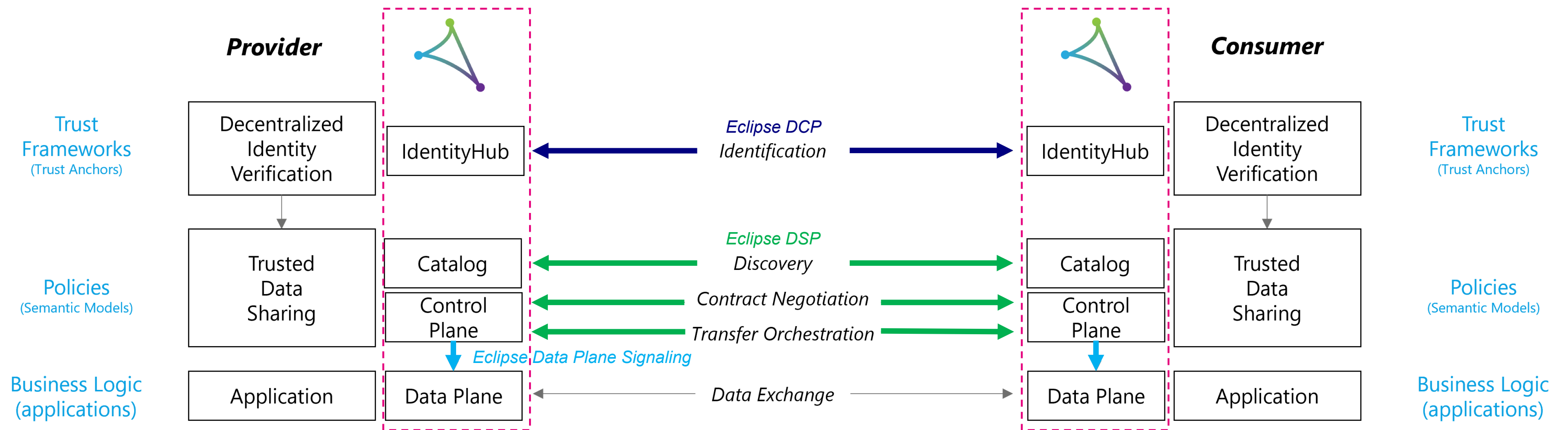
1. The EDC are designed to **serve a range of use cases**, including large AI data sets, API access, supply-chain data processing, and research data sharing
2. The EDC **reuse available technologies** as much as possible. Adopters know better how to optimize for their specific domain
3. The EDC are **standards-based** and implement the Dataspace Protocol Specification and Decentralized Claims Protocol Specification

Architecture Principles

1. **Modular** – Lightweight, composable runtime, minimal dependencies
2. **Extensible** – Replace implementations (e.g., database, security) and create own features and capabilities
3. **Resilient** – Leverages high availability infrastructure (cloud services, data storage, transfer technologies)
4. **Adaptable** – Deploy to diverse environments (cloud, on-premise, edge) with different capabilities

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Leverage Open Source Implementation to Build Dataspaces



Standards are ready, now we need to build something!

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Demo – Open-source components

READMEContributingApache-2.0 licenseSecurity

EDC Samples

This repository provides samples for the Eclipse Dataspace Components (EDC) project, that aim to support the onboarding process for new community members. The samples are sorted by scope for easy navigation and serve as a step-by-step guide for getting familiar with the project. The code and a detailed documentation for each sample reside in the respective module.

Prerequisites

The samples assume a working knowledge of the **EDC nomenclature**. If you do not know about the EDC nomenclature we strongly advise reading the [documentation](#) and/or watching the [introductory videos](#).

Also, a working knowledge of **Git**, **Gradle**, **Java** and **HTTP** is presumed.

We'll assume that you've just checked out the samples code base and have **Java 17+** installed on your development machine. If not, please download and install JDK 17+ for your OS.

Command examples in this document will use the `bash` syntax and use Unix-style paths, but any other shell should be fine as well. If you're using Windows you either need to adapt the paths or use WSL2.

Scopes

The samples are separated by scope, where each scope has a different focus. Depending on what you want to learn more about, you can choose the scope to look at. More scopes may be added in the future, so be sure to check back regularly.

If you are new to the project, it is advisable to start with the `basic` scope. The samples in this scope teach the very basics about using the EDC framework, which are presupposed in all other scopes!

Basic

The samples in this scope teach you how to get started with the EDC framework. You will e.g. learn how to set up and run a connector and how to create your own extensions. What you learn here will be used in all other samples. Click the link above to learn about the basic samples in more detail.

All basic samples are located in the `basic` directory.

Transfer

This scope revolves around the topic of data transfer. In these samples you will learn how a data transfer works in the EDC and run different transfer scenarios. Click the link above to learn about the transfer samples in more detail.

build.gradle.ktsbuild(deps): bump com.bmuschko.docker-remote-api from ...3 months ago

gradle.propertiesIntroduce new snapshot version 0.15.0-SNAPSHOT4 months ago

gradlewIntroduce new snapshot version 0.15.0-SNAPSHOT4 months ago

gradlew.batchore(build): bump Gradle version to 8.102 years ago

seed-k8s.shfeat: seed issuer for database attestations (#447)last year

seed.shchore: fix after upstream changes holder renaming (#446)last year

settings.gradle.ktschore: switch to edc build plugin 1.0.0 (#524)6 months ago

READMEContributingApache-2.0 licenseSecurity

Minimum Viable Dataspace Demo

- Minimum Viable Dataspace Demo
 - 1. Introduction
 - 2. Purpose of this Demo
 - 2.1 Which version should I use?
 - 3. The Scenario
 - 3.1 Participants
 - 3.2 Data setup
 - 3.3 Access control
 - 3.4 DIDs, participant lists and VerifiableCredentials
 - 4. Running the demo (inside IntelliJ)
 - 4.1 Start NGINX
 - 4.2 Starting the runtimes
 - 4.3 Seeding the dataspace
 - 4.4 Next steps
 - 5. Running the Demo (Kubernetes)
 - 5.1 Build the runtime images
 - 5.2 Create the K8S cluster
 - 5.3 Seed the dataspace

kubernetesmvdSearch

Workloads > Deployments

WorkloadsN

Cron Jobs

Daemon Sets

Deployments

Jobs

Pods

Replica Sets

Replication Controllers

Stateful Sets

Service

IngressesN

Ingress Classes

ServicesN

Config and Storage

Config MapsN

Persistent Volume ClaimsN

SecretsN

Storage Classes

Deployments

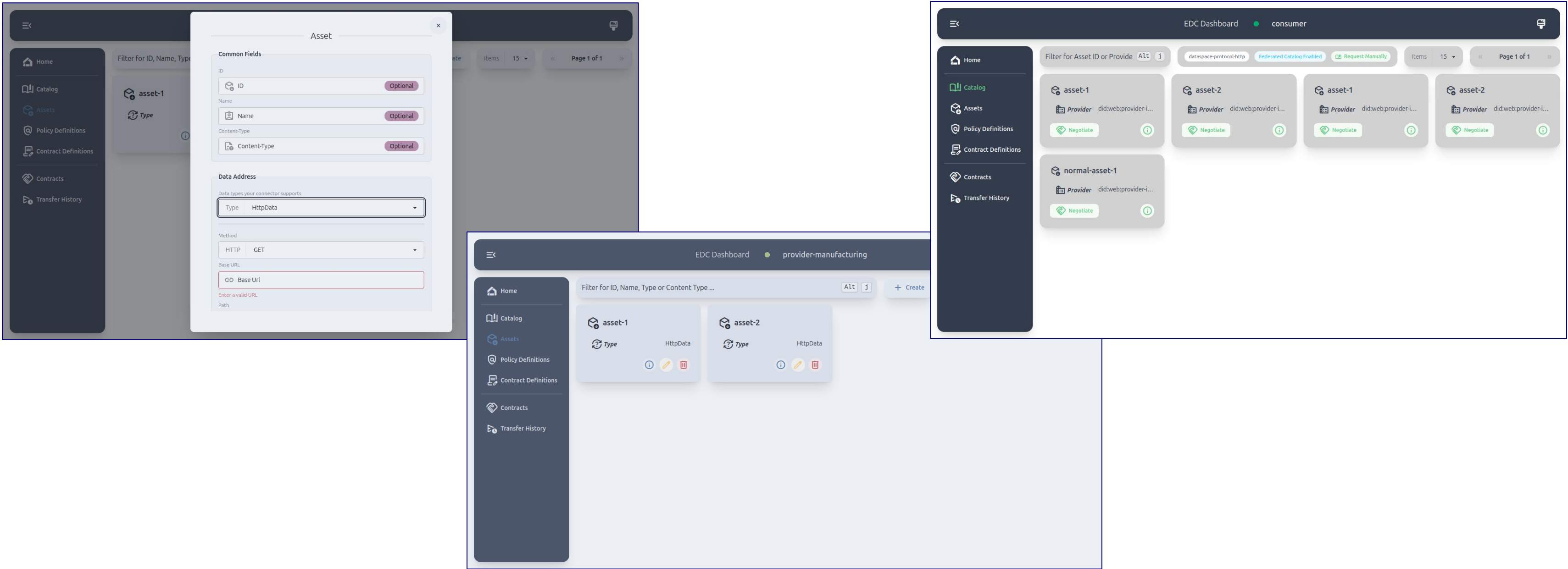
Name	Images	Labels	Pods	Created ↑	
provider-manufacturing-dataplane	dataplane:latest	App: provider-manufacturing-dataplane	1 / 1	17 minutes ago	⋮
provider-qna-dataplane	dataplane:latest	App: provider-qna-dataplane	1 / 1	17 minutes ago	⋮
consumer-dataplane	dataplane:latest	App: consumer-dataplane	1 / 1	17 minutes ago	⋮
consumer-controlplane	controlplane:latest	App: consumer-controlplane	1 / 1	17 minutes ago	⋮
provider-qna-controlplane	controlplane:latest	App: provider-qna-controlplane	1 / 1	17 minutes ago	⋮
provider-catalog-server	catalog-server:latest	App: provider-catalog-server	1 / 1	17 minutes ago	⋮
provider-manufacturing-controlplane	controlplane:latest	App: provider-manufacturing-controlplane	1 / 1	17 minutes ago	⋮
consumer-identityhub	identity-hub:latest	App: consumer-identityhub	1 / 1	18 minutes ago	⋮
dataspace-issuer-service	issuerservice:latest	App: dataspace-issuer-service	1 / 1	18 minutes ago	⋮
provider-identityhub	identity-hub:latest	App: provider-identityhub	1 / 1	18 minutes ago	⋮

1 – 10 of 14<<<>>>

Java 50.0%
HCL 40.5%
Shell 6.6%
Dockerfile 2.0%

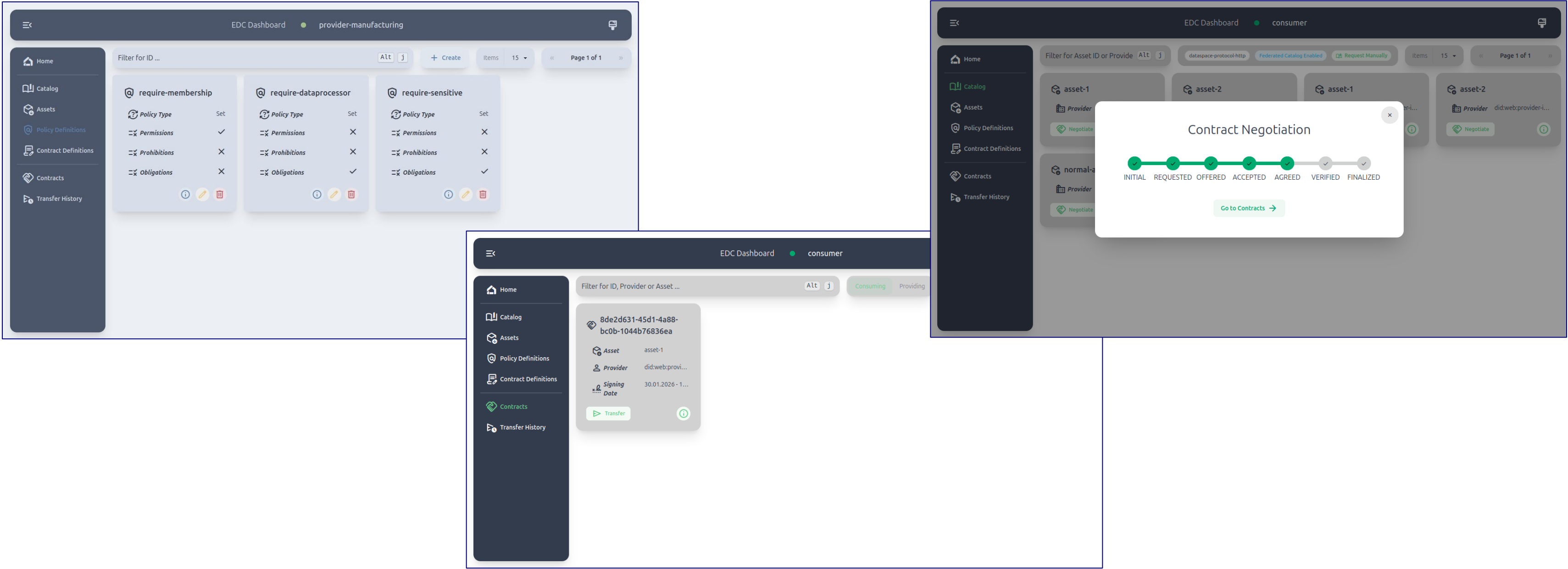
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Demo – Provide & Discover Data



Data Spaces Symposium

Demo – Negotiate & Manage Contracts



Data Spaces Symposium

Demo – Orchestrate & Manage Data Transfers

EDC Dashboard

consumer

Home

Catalog

Assets

Policy Definitions

Contract Definitions

Contracts

Transfer History

Filter for ID, Provider or Asset ...

8de2d631-45d1-4a88-bc0b-1044b76836ea

Asset

Provider

Signing Date

Transfer

Transfer

Contract

8de2d631-45d1-4a88-bc0b-1044b76836ea

Asset

asset-1

Provider

did:web:provider-identityhub%3A7083:provider

Signing Date

30.01.2026 - 11:36:07

Transfer Types offered by the Provider

Select

Transfer Type

HttpData-PULL

Start Transfer

EDC Dashboard

provider-manufacturing

Home

Catalog

Assets

Policy Definitions

Contract Definitions

Contracts

Transfer History

Filter for Transfer ID, State, Transfer Type, Asset ID or Contract ID ...

Consuming

Providing

Items

20

Page 1 of 1

Transfer ID	State Changed	State	Transfer Type	Asset ID	Contract ID
4760ce7e-792f-46df-9de9-57d44a522022	30.01.2026 - 16:39:11	STARTED	HttpData-PULL	asset-1	fd93462d-25ec-41be-8144-2af7d3142e46
33255faa-94a0-47e2-9d5b-375cd28e3bef	30.01.2026 - 16:38:10	STARTED	HttpData-PUSH	asset-1	fd93462d-25ec-41be-8144-2af7d3142e46
5978c19e-dd85-47f5-b7ff-5192e989dd37	30.01.2026 - 16:36:53	STARTED	HttpData-PUSH	asset-1	fd93462d-25ec-41be-8144-2af7d3142e46
1199ffad-c8f7-463e-a20a-d5b2842cacf8	30.01.2026 - 16:28:33	STARTED	HttpData-PULL	asset-1	fd93462d-25ec-41be-8144-2af7d3142e46

EDC Dashboard

consumer

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Catalog

Assets

Policy Definitions

Contract Definitions

Contracts

Transfer History

Filter for ID, Provider or Asset ...

8de2d631-45d1-4a88-bc0b-1044b76836ea

Asset

Provider

Signing Date

Transfer

Pull Transfer

INITIAL

PROVISIONED

REQUESTED

READY

Data Spaces Symposium

Let's Build Dataspaces!

How to contribute?

- GitHub: developers, technical writers, designer, ...
- Eclipse Dataspace Working Group (EDWG)
- Write your own extension(s)
- Sponsor the work through budget



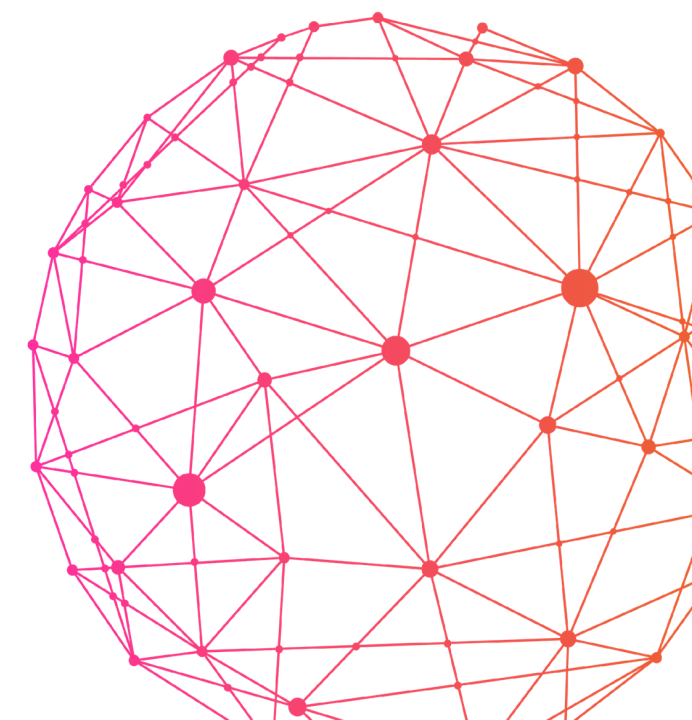
GitHub



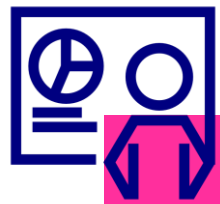
Getting Started
(Samples)



MVD
(Demo Setup)



Explore EDC @ Data Spaces Symposium



Join the Breakout Session

**“Who onboards the millions?
Solving data spaces at scale”**



Madrid



February 11



11 a.m.



Talk about the EDC at the booths

- Eclipse
- Data Space Adoption Forum
- PwC



**Data Space
Adoption Forum**

